

MD. RAFIUL BISWAS, PhD

Postdoctoral Researcher | NLP • Multimodal AI • Arabic Content Safety

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EDUCATION

PhD in Computer Science and Engineering

August 2020 – May 2024

Hamad Bin Khalifa University (HBKU), Doha, Qatar

- Dissertation: "Identifying Determinants of Vaccine Hesitancy from Twitter Data"
- Advisors: Dr. Zubair Shah (HBKU), Dr. Raian Ali (HBKU)

MSc in Computer Science and Engineering

January 2017 – September 2018

Khulna University of Engineering & Technology, Bangladesh

BSc in Computer Science and Engineering

January 2012 – May 2016

Khulna University of Engineering & Technology, Bangladesh

RESEARCH INTERESTS

Hate speech detection, offensive content classification, and multimodal AI for low-resource languages. Specific areas: multi-label hate speech, propaganda and persuasion identification, hateful meme detection (vision-language models), emotion and stance in polarized discourse, Arabic NLP and LLM adaptation, content safety benchmarking for Arabic dialects.

ACADEMIC & RESEARCH EXPERIENCE

Postdoctoral Researcher

August 2024 – Present

Hamad Bin Khalifa University (HBKU) & Northwestern University in Qatar (NU-Q) | Doha, Qatar

- Lead development of MARSAD—a production-scale platform for real-time monitoring and classification of harmful content (hate speech, propaganda, polarized stance) across multiple social platforms.
- Co-organize ArGuard 2026, MAHED 2025, StanceNakba 2026, and CheckThat! 2025; mentor junior researchers and supervise MSc projects.
- Publish in top NLP venues (ACL, EMNLP, LREC, WWW); deliver invited talks and present at international conferences.
- 50+ peer-reviewed publications; 800+ citations.

Graduate Teaching Assistant

February 2021 – February 2024

Hamad Bin Khalifa University | Doha, Qatar

- Taught Principles of Computing (Python) and Information Management Systems (graduate-level).
- Mentored MSc students in research methodology, data analysis, and academic writing.

Software Engineer

January 2019 – August 2020

Cranium Soft Ltd | Bangladesh

- Designed and developed full-stack enterprise software for higher-education strategic planning (ASP.NET, Java).

SELECTED RESEARCH PROJECTS

ArGuard 2026 — Two-Track Benchmark for Arabic Content Safety

Co-Organizer | araieval.github.io/ArGuard2026

- Co-organizing a two-track benchmark for Arabic content safety combining multimodal hateful meme understanding with harmful prompt detection for LLM interactions.

- Evaluates both memes (visual cues + embedded text) and unsafe LLM prompts under a shared design moving from binary detection to fine-grained categorization.
- Built on ArabicDialectSafety, a dialect-aware benchmark for Arabic content safety classification across six dialect varieties (arXiv:2608.01291).

MAHED 2025 — Multimodal Detection of Hope and Hate Emotions in Arabic

Lead Co-Organizer | ArabicNLP 2025 (co-located with EMNLP)

- Organized shared task with 46+ team submissions across three subtasks: text-based hate/hope classification, multi-task emotion+offensive+hate detection, and multimodal hateful meme detection.
- Curated and released 22,000+ annotated instances from Twitter and Facebook; inter-annotator agreement Cohen's $\kappa > 0.85$.
- Benchmarked AraBERT, MARBERT, GPT-4, Gemini, and CLIP-Arabic multimodal fusion architectures; published overview paper at ACL Anthology.

MARSAD — Real-Time Social Media Observatory

Lead Developer | Hamad Bin Khalifa University, 2024–Present

- Architected production-scale platform for real-time harmful content detection across multiple social platforms.
- Full ML pipeline: PyTorch backend, transformer-based classifiers, OCR-multimodal fusion, real-time inference, and interactive analyst dashboard.
- Serves as research infrastructure for ongoing studies on online harm at population scale.

Multi-Label Arabic Hate Speech Corpus

Lead Researcher | LREC-COLING 2024

- Built one of the largest fine-grained multi-label hate speech datasets for Arabic, annotated with hate targets, emotions, and offensive attributes.
- Annotation guidelines now adopted by downstream shared tasks; collaborating with Hamdy Mubarak (QCRI) on quality assurance.

StanceNakba 2026 — Actor and Topic-Aware Stance Detection

Co-Organizer | LREC 2026 (Nakba-NLP Workshop)

- Dual-framework shared task on polarized discourse: actor-level stance detection (English) and cross-topic stance detection (Arabic).
- Released 2,606 manually annotated social media posts with rigorous quality control.
- Website: marsadlab.github.io/stancenakba

Multimodal Propaganda & Hate Detection in Arabic Memes

Co-Investigator | ArAIEval 2024 + WISE 2024

- Developed vision-language models using LLaMA-3.2-Vision, Qwen-VL, and CLIP-based architectures for multimodal hate and propaganda detection.
- Introduced multi-agent LLM framework for fine-grained meme analysis ("Propaganda to Hate," Springer, WISE 2024).

SELECTED PUBLICATIONS

** indicates corresponding/lead author*

2026

- Zaghouani, W., Biswas, M.R.*, Aldous, K.K., & Bessghaier, M. "ArabicDialectSafety: A Dialect-Aware Benchmark for Arabic Content Safety Classification." *arXiv:2608.01291*.
- Kmainasi, M.B., Shahroor, A.E., Hasnat, A., Biswas, M.R., Zaghouani, W., & Alam, F. "AHA-Memes: A Fine-Grained Multimodal Benchmark for Understanding Hate in Arabic Memes." *arXiv:2607.27393*.

- Aldous, K.K., Biswas, M.R.*, Bessghaier, M., Ibrahim, S., Attia, K., & Zaghouani, W. "StanceNakba Shared Task: Actor and Topic-Aware Stance Detection in Public Discourse." *arXiv:2606.12068*.
- Zaghouani, W., Attia, K., Biswas, M.R., & Eryani, F. "ArPoMeme: An Annotated Arabic Multimodal Dataset for Political Ideology and Polarization." *arXiv:2605.20967*.
- Zaghouani, W., Biswas, M.R., Bessghaier, M., Ibrahim, S., & Mikros, G. "ClimateChat-300K: A Multi-Modal Facebook Dataset for Understanding Diverse Perspectives in Climate Communication." *arXiv:2605.23326*.
- Islam, A., Biswas, M.R., Belhaouari, S.B., & Zaghouani, W. "PushingBoundaries at NakbaVirality Shared Task: Recursive Prompt Improvement for Multimodal Virality Classification." *Proceedings of the 2nd International Workshop on Nakba Narratives, 2026*.
- Biswas, M.R.*, Bessghaier, M., Ibrahim, S., Mikros, G., & Zaghouani, W. "Longitudinal Trends in Global Climate Change Discourse on Facebook." *Proceedings of the ACM Web Conference (WWW '26), pp. 9473–9481*.
- Bessghaier, M., Biswas, M.R., Ibrahim, S., & Zaghouani, W. "A Multi-Task Learning Framework for Modeling Engagement and Topic-Sensitive Responses in Arabic Women's Discourse." *Findings of the Association for Computational Linguistics: EACL 2026, pp. 4846–4854*.
- Khan, S., Mohamed, S.P., Biswas, M.R., & Shah, Z. "Exploring the Capabilities of Large Language Models in Oral and Maxillofacial Surgery." *Measurement and Control, 59(5), pp. 640–649*.

2025

- Zaghouani, W., Biswas, M.R.*, Bessghaier, M., Ibrahim, S., Mikros, G., Hasnat, A., & Alam, F. "MAHED Shared Task: Multimodal Detection of Hope and Hate Emotions in Arabic Content." *ArabicNLP 2025 (ACL Anthology), pp. 560–574*.
- Zaghouani, W., & Biswas, M.R. "An Annotated Corpus of Arabic Tweets for Hate Speech Analysis." *RANLP 2025, pp. 1413–1419*.
- Zaghouani, W., & Biswas, M.R. "EmoHopeSpeech: An Annotated Dataset of Emotions and Hope Speech in English and Arabic." *RANLP 2025, pp. 1406–1412*.
- Biswas, M.R.*, Alam, F., & Zaghouani, W. "MARSAD: A Multi-Functional Tool for Real-Time Social Media Analysis." *arXiv:2512.01369*.
- Biswas, M.R.*, Ibrahim, S., Bessghaier, M., & Zaghouani, W. "Evaluation of Pretrained and Instruction-Based Models for Emotion Detection in Arabic Social Media Text." *RANLP 2025*.

2024

- Zaghouani, W., Mubarak, H., & Biswas, M.R. "So Hateful! Building a Multi-Label Hate Speech Annotated Arabic Dataset." *LREC-COLING 2024, pp. 15044–15055*.
- Alam, F., Biswas, M.R., Shah, U., Zaghouani, W., & Mikros, G. "Propaganda to Hate: A Multimodal Analysis of Arabic Memes with Multi-Agent LLMs." *WISE 2024, Springer, pp. 380–390*.
- Shah, U., Biswas, M.R., Agus, M., Househ, M., & Zaghouani, W. "MemeMind at ArAIEval: Generative Augmentation and Feature Fusion for Multimodal Propaganda Detection in Arabic Memes." *ArabicNLP 2024, pp. 467–472*.
- Hasanain, M., Hasan, M.A., Ahmad, F., Suwaileh, R., Biswas, M.R., Zaghouani, W., & Alam, F. "ArAIEval Shared Task: Propagandistic Techniques Detection in Unimodal and Multimodal Arabic Content." *ArabicNLP 2024, pp. 456–466*.
- Biswas, M.R.*, Shah, Z., & Zaghouani, W. "MemeMind at ArAIEval Shared Task: Spotting Persuasive Spans in Arabic Text with Persuasion Techniques Identification." *ArabicNLP 2024 (ACL), pp. 494–500*.
- Biswas, M.R.*, Abir, A.T., & Zaghouani, W. "Nullpointer at CheckThat! 2024: Identifying Subjectivity from Multilingual Text Sequence." *arXiv:2407.10252*.

SHARED TASKS & ACADEMIC SERVICE

Shared Task Organization

- **ArGuard 2026 — Co-Organizer:** Two-track Arabic content safety benchmark: multimodal hateful memes + harmful LLM prompts
- **MAHED 2025 — Lead Co-Organizer:** Multimodal hate and hope emotion detection in Arabic (46+ teams, ArabicNLP @ EMNLP)
- **StanceNakba 2026 — Co-Organizer:** Actor and topic-aware stance detection in Palestinian-Israeli discourse (LREC 2026)
- **ArAIEval 2024 — Co-Organizer:** Multimodal propaganda detection in Arabic memes (ArabicNLP @ ACL)
- **CheckThat! 2025 (CLEF) — Co-Organizer:** Multilingual subjectivity and check-worthiness detection
- **ImageEval 2025 — Co-Organizer:** First Arabic image captioning shared task

Peer Review

- Journals: IEEE Internet of Things Journal, IEEE JBHI, ACM TALLIP, Springer Nature, Digital Health
- Conferences: ACL, EMNLP, LREC (ArabicNLP track, expert reviewer)

TECHNICAL EXPERTISE

- **NLP & LLMs:** HuggingFace Transformers, AraBERT, MARBERT, AceGPT, JAIS, FANAR, ALLaM, Falcon, GPT-4, LLaMA, Qwen
- **Multimodal Models:** CLIP-ViT, LLaMA-3.2-Vision, Qwen-VL, MBZUAI/AIN, multimodal fusion, OCR-aware architectures
- **Deep Learning:** PyTorch, TensorFlow/Keras, LoRA/QLoRA fine-tuning, instruction tuning, multi-task learning, knowledge distillation, DeepSpeed, vLLM
- **Data & Annotation:** Annotation pipeline design, inter-annotator agreement (Cohen's K), dataset curation, social media collection, evaluation benchmarks, Codabench leaderboards
- **Infrastructure:** SLURM/HPC clusters, Northwestern Quest HPC, Docker, Git, REST APIs; Python (expert), Java, JavaScript, ASP.NET

AWARDS & RECOGNITION

- **CodeCamp 2022** — 25,000 QAR Cash Award + 2-year Incubation, Digital Incubation Center (MCIT, Qatar)
- **Al-Fikra 2022 National Entrepreneurship Competition** — 15,000 QAR + 50,000 QAR Incubation Fund (Qatar Development Bank & Qatar University)
- **HBKU Student Innovation Grant 2023** — Shariah Advisor Chatbot project
- **HBKU Innovation Grant 2023** — Corporate Social Responsibility Through Sports
- Microsoft Certified: Azure AI Engineer Associate | IBM AI Analyst Mastery Award

LANGUAGES

English (Proficient) • Bengali (Native) • Arabic (Intermediate)

REFERENCES

Dr. Wajdi Zaghouani wajdi.zaghouani@northwestern.edu
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Dr. Zubair Shahzshah zubair.shahzshah@hbku.edu.qa
Assistant Professor, Hamad Bin Khalifa University

Dr. Firoj Alam Available upon request
Scientist, Qatar Computing Research Institute (QCRI)